

```

import itertools
import time

def brute_force_pin(actual_pin):
    attempts = 0
    start_time = time.time()

    # Generate all possible 4-digit combinations (0000 to 9999)
    for combo in itertools.product(range(10), repeat=4):
        attempts += 1
        guess = ''.join(map(str, combo))

        if guess == actual_pin:
            elapsed = time.time() - start_time
            print(f"PIN found: {guess}")
            print(f"Number of attempts: {attempts}")
            print(f"Time taken: {elapsed:.4f} seconds")
            return guess, attempts

    print("PIN not found (this shouldn't happen for a 4-digit PIN).")
    return None, attempts

if __name__ == "__main__":
    # Preset PIN to guess (as a string to preserve leading zeros)
    actual_pin = "4271"

    print(f"Simulating brute-force attack on a 4-digit PIN...")
    brute_force_pin(actual_pin)

```

OUTPUT:

```

Simulating brute-force attack on a 4-digit PIN...
PIN found: 4271
Number of attempts: 4272
Time taken: 0.0031 seconds

```